

Comparison of laboratory and field environmental conditioning on FRP-concrete bond durability



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HIGHLIGHTS

- CFRP-concrete bond was evaluated following field and accelerated conditioning (AC).
- Direct tension pull-off and notched three-point bending tests were utilized to test the bond.
- AC resulted in degradation of bond properties.
- Some field conditioned samples experienced slight degradation between 6 and 18 months.
- ACP at 60 °C may result in a pessimistic estimate of bond durability for short service life.

ARTICLE INFO

Article history:

Received 24 November 2015
Received in revised form 3 June 2016
Accepted 14 June 2016
Available online 9 July 2016

Keywords:

FRP
Concrete
Adhesive
Epoxy
Bond
Durability
Field
Accelerated conditioning
Degradation
Bridge engineering

ABSTRACT

Assessment of bonded FRP durability by means of accelerated conditioning is sometimes thought to be too harsh compared to ambient environmental conditions; consequently, this may result in underestimation of the actual durability. To assess the efficacy of utilizing accelerated conditioning protocols (ACP) for FRP-concrete bond durability testing, results from laboratory and field conditions (Sunshine Skyway Bridge in Tampa, FL) were compared. Direct tension pull-off test patches were applied to the approach span girders and notched beam three-point bending test specimens were placed on the dolphins adjacent to the bridge. These results were compared to notched beam specimens exposed to ACP. Testing indicated that characteristics of the FRP-concrete bond failure modes changed in some of the field samples within 6 months of field exposure, which may be an indication of durability problems. Moreover, ACP under elevated temperatures (60 °C) of notched three-point bending test resulted in a 36% loss of strength compared to no strength degradation after 18 months of field conditioning.

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1. Introduction

FRP composites have been used extensively in the past 20 years to retrofit bridges in need of repair or strengthening. The severity of the environmental conditions experienced by externally bonded FRP reinforcement in such applications can affect the durability of the adhesive joint between FRP and concrete. Accelerated conditioning protocols (ACP) are commonly used to assess the durability by accelerating the aging process of the bimaterial system in a laboratory setting; the applicability and relationship of ACP to in-situ, real-time environmental conditions, however, is not fully understood.

This paper presents a study that compares the effect of ACP to that of real-time exposure on the FRP bond properties. The bond properties were tested using notched three-point bending and direct tension pull-off test methods. Field conditioning was conducted on the Sunshine Skyway Bridge in Tampa, FL, which had recently been repaired with a commercially available wet layup bonded FRP composite system. This same system was applied to small beam specimens, which were then placed on the bridge dolphins. Companion beam specimens were also subjected to accelerated conditioning protocols (ACP) for comparison.



2. Literature review

Multiple authors have reported results of durability tests on FRP-concrete bond, concentrating primarily on the effects of dry

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heat [17], moisture [9], salt and moisture [18], freeze-thaw cycles [13], and alkaline environments [23], among others. The agreement is mutual, however, that the most critical environment for FRP-concrete bonded joints is moisture. Choice of conditioning temperature in hygrothermal environments is the most challenging aspect of selecting an appropriate accelerated conditioning protocol; accelerating degradation by using temperatures higher than the glass transition temperature (T_g) of the epoxy adhesive used to form the composite and/or the bond between composite and concrete substrate can introduce degradation mechanisms that may not accurately represent those occurring in the field under real time.

AASHTO [4], International Code Council Evaluation Service (ICC-ES) [16], and ACI [7], however, recommend multiple conditioning protocols that specify a temperature of 60 °C. While this temperature can exceed the T_g of the composite-saturating resin or adhesive, conditioning at temperatures slightly higher than T_g is thought to be beneficial to establishing a lower bound estimate of residual strength. On the other hand, ACI [8] prescribes a conditioning temperature of 50 °C (slightly lower than the T_g of most commercial epoxies) to avoid introducing the glass transition during the conditioning. While convenient, it is not clear whether testing at temperatures higher than T_g is justified. The research reported in this paper compares the field durability data to those obtained in the laboratory and draws conclusions based on the findings.

3. Experimental procedures

Field exposure and ACP were compared by measuring the bond strength retention using the direct tension pull-off test method, and the notched beam three-point bending test method, which are detailed in the following sections.

3.1. Three-point bending test method

The design and test procedures followed for three-point bending test were implemented from Gartner et al. [12]. Concrete specimens had a square cross section measuring 10 cm by 10 cm and were 35 cm long (Fig. 1). The amount of FRP reinforcement was chosen to allow for a debonding failure mode. Higher reinforcement ratios usually result in flexure-shear failure of the test specimen, which does not provide an assessment of the bond durability; FRP strips measuring 2.5 cm in width and 20 cm in length achieved the desired failure mode. To better represent

the cracked concrete, a 5 cm deep notch was introduced at the specimen midspan. The notch also eliminated the need for a constant moment region (debonding path is predefined), allowing for use of three-point bending instead of four-point bending loading condition.

The concrete mixture used to produce the concrete beam specimens had a target compressive design strength of 69.9 MPa, which was accomplished by using a low water to cementitious material ratio (w/cm) of 0.353. The cement used was Type I/II (ASTM C150 [1]), and the ratio of cement:sand:coarse aggregate was 1:1.74:2.22 by volume. The coarse aggregate with maximum size of 9 mm and the gradation conforming to #89 was used (ASTM C33 [2]). The mix had a slump of 64 mm and air content of 3.5%. The actual 28-day compressive strength was 78.7 MPa.

Test specimens were loaded at a constant displacement rate of 0.04 mm/min. until the ultimate failure. All specimens experienced a debonding failure mode (Fig. 2). The loading rate was selected to induce an increase in average FRP bond stress between 0.4 and 0.8 MPa/min [15], which generally resulted in total time to failure of less than three minutes; this loading rate is believed to minimize the effects of creep of the epoxy adhesive and is in conformance with ACI 440.9R-15 guidelines.

Test results from the three-point bending test are presented as bond strength retention (R_b):

$$R_b = \frac{\text{ACP strength}}{\text{SLC strength}} \quad (1)$$

where “ACP strength” is the mean ultimate strength of beams exposed to the selected (ACP) and “SLC strength” is the mean ultimate strength of beams exposed to standard laboratory conditions (SLC), as defined in ACI 440.9R-15 (temperature of 23 ± 3 °C, and relative humidity of $50 \pm 10\%$).

3.2. Pull-off test method for FRP-concrete bond

Direct tension pull-off test (Fig. 3) is specified under ASTM D7522 [3]. The test is most commonly utilized as a part of quality control in newly installed repairs with externally bonded FRP. In this study, direct pull-off testing was conducted to assess the durability of FRP-concrete adhesive bond in-situ and in the laboratory setting. One side of the three-point bending test specimen had a FRP patch for the purpose of pull-off bond testing (Fig. 2). Results are presented in terms of R_b , in a similar manner as the notched beam three-point bending test.

4. Composite system

Durability of bond of one wet-layup CFRP composite system was evaluated (Composite A). Composite A system consists of a low-viscosity polyamine epoxy primer, paste epoxy putty, and dry carbon fiber fabric that, when soaked with the saturant, forms

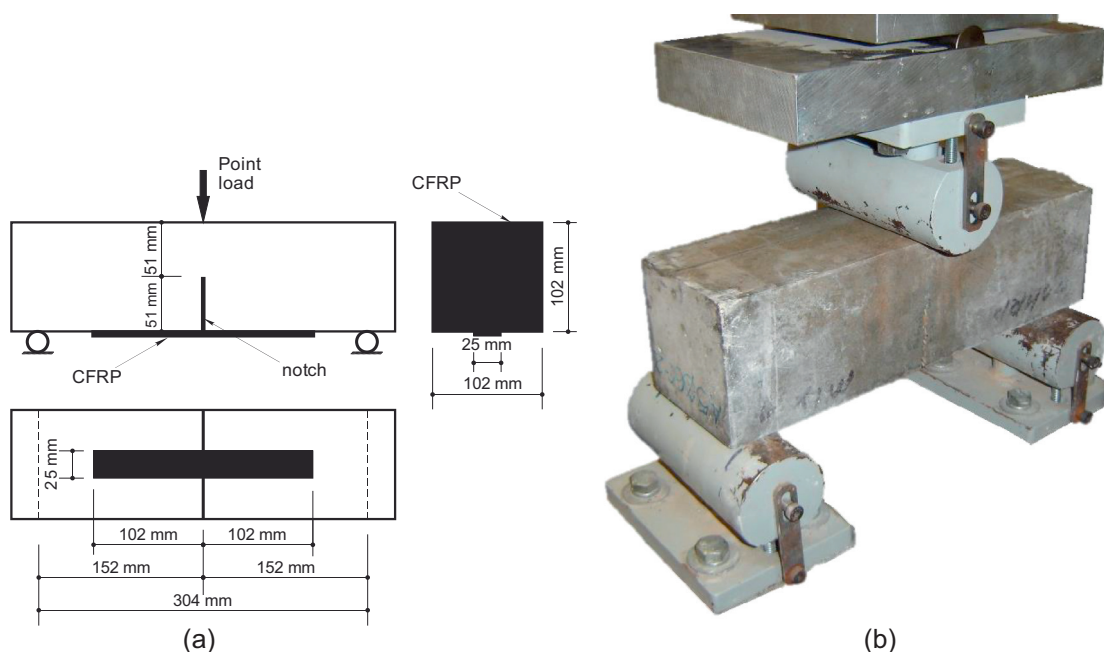


Fig. 1. (a) Notched three-point bending test schematic; and (b) test setup.



Fig. 2. Debonding failure mode.

a CFRP composite. Composite A mechanical properties, as reported by the manufacturer, are summarized in Table 1.

Properties of different epoxy components used as a part of composite A are shown in Table 2. The differences between the glass transition temperature (T_g) values reported by the manufacturer and those reported by Blackburn et al. [10] are likely due to different test methods used, reporting different temperature in the glass transition range, and most importantly different curing histories of epoxy samples prior to testing. The results reported by Blackburn et al. [10] correspond to 12 weeks of curing in SLC.

5. Laboratory accelerated conditioning

By utilizing the accelerated conditioning protocols (ACP) recommended by Stewart [19], assessment of durability of bond between Composite A and concrete was conducted by means of notched three-point bending test (Fig. 1) and direct tension pull-off testing (Fig. 3). The specimens were tested at 1, 2 and 8 weeks following the exposure to: (1) water immersion at 30 °C; (2) water immersion at 60 °C; and (3) RH = 100% at 60 °C. Control specimens were kept in SLC throughout the exposure, and tested together with the 8-week-conditioning samples. All exposed specimens were tested in three point-bending within the 24 h of exposure. Direct tension pull-off testing was conducted on 8-week samples only, at a later time. Three specimens were tested in each test group in addition to three control specimens for a total of 30 specimens.

6. Field exposure

Composite A was applied to the approach span girders to alleviate the effects of web cracking near the supports on the Sunshine

Table 1
Composite A mechanical properties.

Property	Value
Dry fiber weight	600 g/m ²
Elastic modulus	227 GPa
Tensile strength	3800 MPa
Elongation	1.67%

Table 2
Composite A epoxy components.

Epoxy	Description	Elastic modulus (GPa)	Tensile strength (MPa)	T_g^a (°C)	T_g^b (°C)	Conversion ^b
Primer	Clear epoxy	0.72	17.2	77	N.	N.S.
Putty	Paste epoxy	1.80	15.2	75	45	N.S.
Saturant	Opaque epoxy	3.03	55.2	71	57	0.97

N.S.: not specified.

^a Specified by manufacturer.

^b Measured by Blackburn et al. [10].

Skyway Bridge in Tampa Bay, Florida; multiple locations were wrapped with CFRP strips as shown in Fig. 4a. For the purpose of this study, at the time of the repair, four Composite A test patches measuring 60 × 60 cm were placed on the approach span girders at the north abutment. The patches were installed as per manufacturer's instructions at previously sandblasted areas on both the interior (Fig. 4b) and exterior girders (Fig. 4c). As shown in Fig. 4, a half of each patch was coated with acrylic UV coating; two halves of each patch were separated vertically to account for the possible effects of UV coating on rainfall. Interior and exterior girder locations were chosen to cover a range of environmental conditions experienced by CFRP-concrete bond during its service life. Exterior girder location was deemed to be exposed to a harsher combination of UV radiation and rainfall when compared to the interior girder location. To assess the durability of bond between the Composite A and concrete, direct tension pull-off tests were performed at 6 and 18 months of exposure.

Archival records indicate the girder concrete of the Sunshine Skyway approach span had a 28-day compressive strength of 50.6 MPa. The w/cm ratio of the concrete mix was 0.34; maximum slump was 76 mm, and the air content was measured to be 3.5%. The course aggregate consisted of 25-mm maximum size and

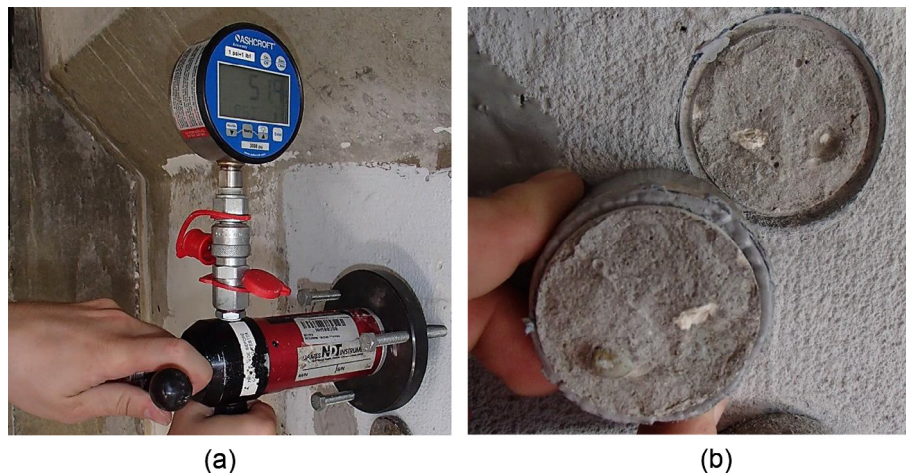


Fig. 3. (a) Pull-off test setup; and (b) failure surface after pull-off test.

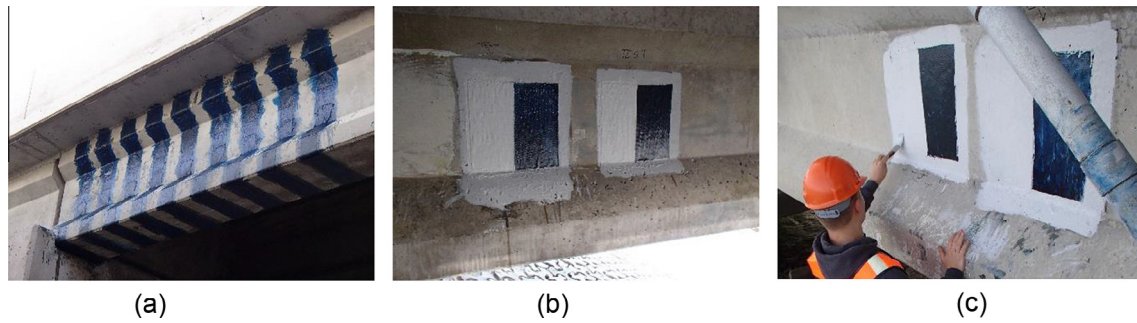


Fig. 4. (a) CFRP strips on the Sunshine Skyway approach span girder; (b) interior girder test patches; and (c) exterior girder test patches.

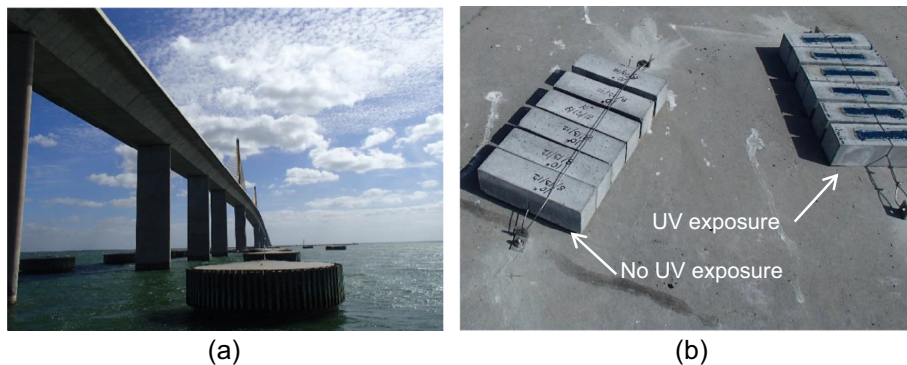


Fig. 5. (a) Sunshine Skyway Bridge dolphin; and (b) specimens on Sunshine Skyway Bridge dolphin.

conformed to a #57 gradation (ASTM C33 [2]). Type II cement was used (ASTM C150 [1]); the ratio of cement:sand:coarse aggregate was 1:1.01:1.88 by volume.

A set of notched three-point bending test specimens were placed on the Sunshine Skyway dolphin structure (Fig. 5a). While this environment is realistic, it is thought to provide slightly accelerated environmental conditions than those found on the bridge; UV radiation, wave, spray, and diurnal temperature changes are deemed harsher. To allow for in-situ curing conditions specimens were prepared in the laboratory and deployed to the dolphin within the 48-h time window. Three distinct exposure test groups were formed: (1) UV exposure without UV coating; (2) UV exposure with UV coating; and (3) no UV exposure with UV coating. To allow for UV exposure the specimens were placed top-down (Fig. 5b); the specimens without UV exposure were put on plywood pads with FRP facing the ground (Fig. 5b). All specimens were secured from wind and wave action using a stainless steel rope. Specimens were collected from the dolphin at 6 and 18 month of exposure. The testing was conducted at the University of Florida structural laboratory in Gainesville, FL within the 24 h of collection from the dolphin. Six specimens were tested in each test group.

7. Results and discussion-laboratory accelerated conditioning

The behavior of each notched three-point bending specimen conformed to that described by Gartner et al. [12]. The average coefficient of variation (COV) of all test series was approximately 7.50% reflecting good repeatability of the test results. Analysis of the durability test data showed the sensitivity of Composite A system to the conditioning temperature (Fig. 6). Comparatively higher R_b values were observed at 1 week of exposure in 60 °C samples, likely because high temperature acted as a catalyst for post-cure of the epoxy adhesive. At 8 weeks of exposure, however,

significantly lower R_b values were observed in 60 °C samples when compared to those conditioned at 30 °C.

In each failed three-point bending test specimen the failure mode was visually evaluated and classified. Classification was expressed as a portion of bonded surface that failed adhesively in percentages, in increments of 5%. Example of such classification is shown in Fig. 7. It is expected that adhesive failure mode characteristics would increase with the severity and duration of exposure. Failure mode of the specimens conformed to the measured R_b values—adhesive failure mode characteristics were more prominent in the 60 °C samples, particularly at 8 weeks of conditioning (Fig. 6b).

Similarly to notched beam three-point bending test, R_b values from direct tension pull-off test showed decreasing trends for all ACP, with the highest degradation observed in specimens conditioned by water immersion at 60 °C (Fig. 8a). Change in observed failure mode between control and exposed specimens was more pronounced than in the three-point bending tests (Fig. 8b); a complete transition from fully cohesive to adhesive failure mode was observed for conditioning at 60 °C, with the adhesive failure mode characteristics being more distinct in specimens conditioned at 60 °C. Examples of visual failure mode classification are given in Fig. 9.

Before comparison of field and AC exposure conditions can be made, the relationship between direct tension pull-off and beam testing must be defined. To determine the correlation between R_b corresponding to direct tension pull-off and notched beam three-point bending tests a bivariate kernel distribution estimate was generated for the laboratory test data (Fig. 10a). By assuming the statistical independence between the two tests, the density (z-axis) was calculated by multiplying the two probability density functions. The resulting plots show no correlation between the two sets of data (Fig. 10b); corresponding covariance and correlation coefficient were determined to be 0.012 and 0.109, respectively.

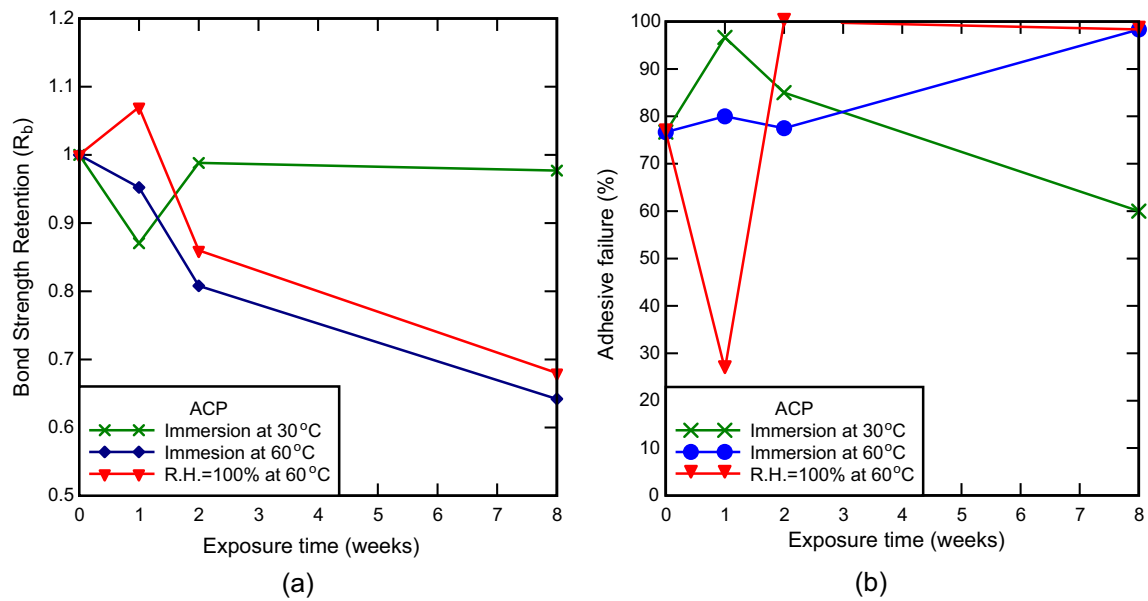


Fig. 6. Results from notched beam three-point bending test: (a) change in R_b ; (b) development of adhesive failure mode.

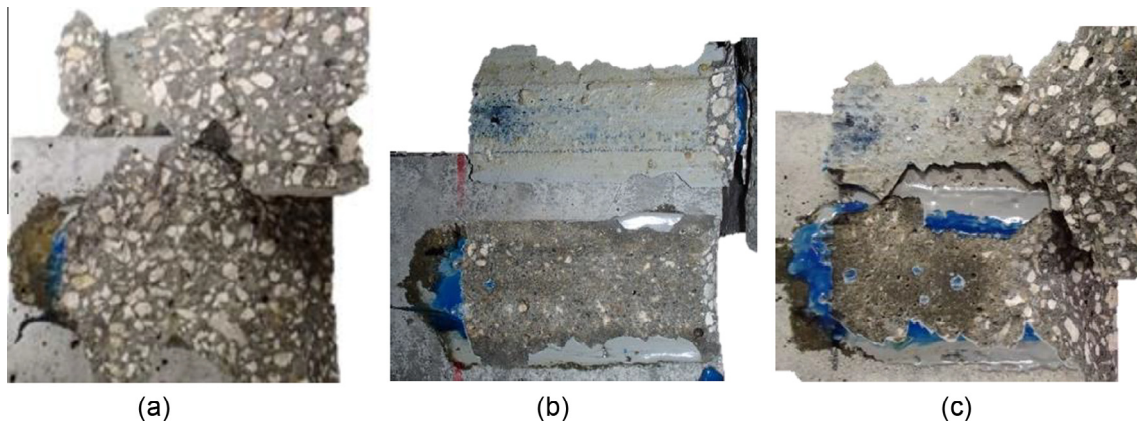


Fig. 7. Classification of failure modes from notched beam three-point bending test: (a) cohesive (0% adhesive); (b) 100% adhesive; and (c) 70% adhesive.

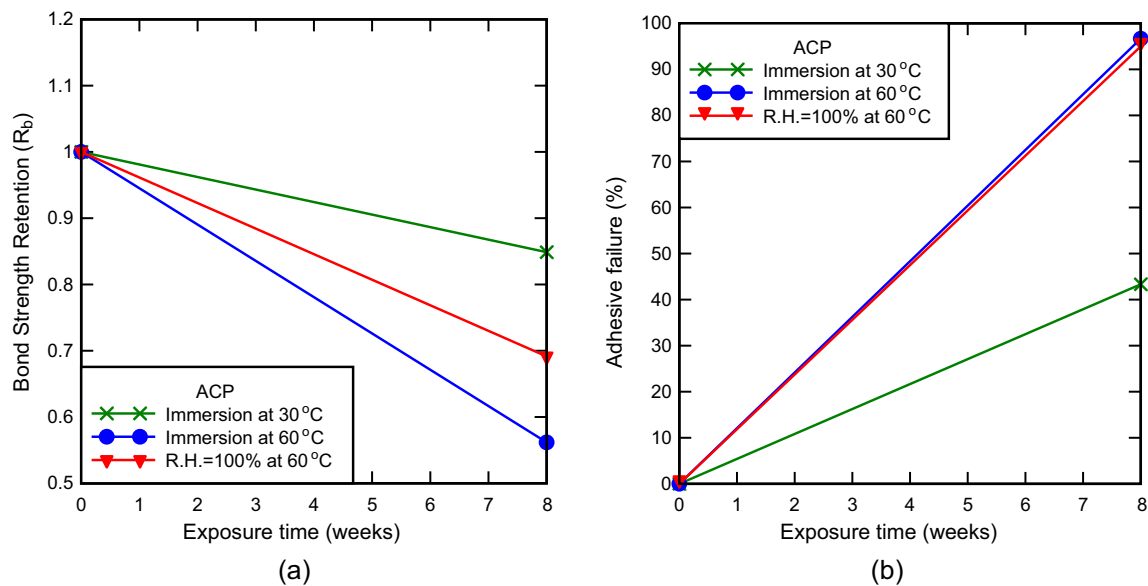


Fig. 8. Results from direct tension pull-off test: (a) change in R_b ; (b) development of adhesive failure mode.

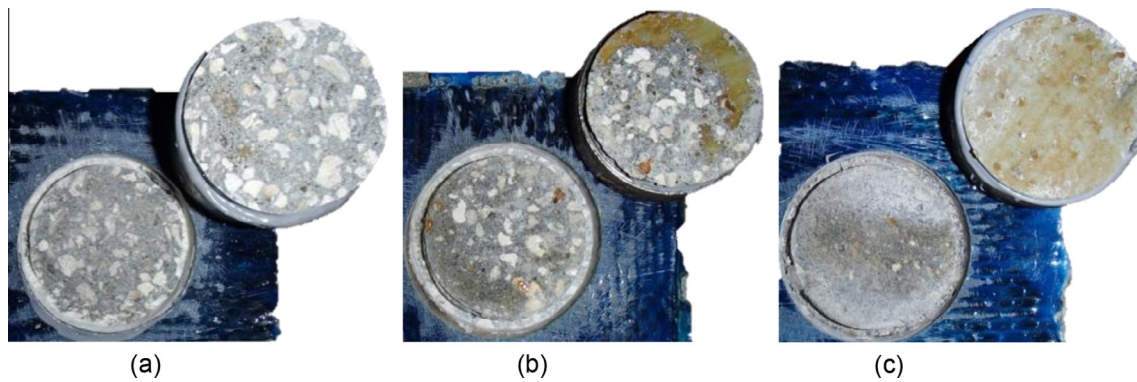


Fig. 9. Classification of failure modes from direct tension pull-off test: (a) cohesive (0% adhesive); (b) 30% adhesive; and (c) 100% adhesive.

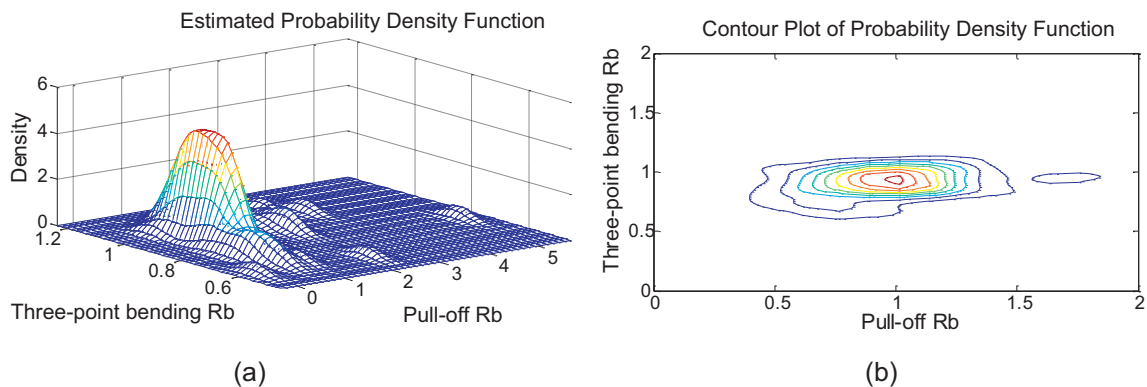


Fig. 10. (a) Estimated multivariate probability density function; and (b) contour plot of estimated probability density function.

The apparent independence of the two test methods can be explained by the mechanics of the two loading modes. Generally, adhesive bond to concrete is comprised of some combination of mechanical interlock and chemical bond [20]. In Mode I loading (direct tension pull-off test) mechanical interlock does not contribute significantly to the bond strength (Fig. 11). In Mode II loading, however, (notched beam three-point bending test) mechanical interlock is an important factor influencing the ultimate strength of the test specimen (Fig. 12). Differences also exist in the repeatability of the two test methods, which are reflected in the elongated shape of the contour diagram (Fig. 10). It is apparent that direct pull-off test has a significantly larger scatter of obtained test values when compared to notched beam three-point bending test method.

In conclusion, while both test methods purport to test bond, they each activate related but dissimilar failure mechanisms. Both are useful in evaluating bond strength and durability; care must be taken, however, when comparing test results between the methods. Furthermore, it is inadvisable to use the data interchangeably. Additional research is required to unravel the complexity of bonding mechanisms and their degradation.

8. Results and discussion-field exposure

8.1. Pull-off tests

Testing on the girder FRP patches was conducted after 6 and 18 months of exposure. At each time interval, four direct tension pull-off tests were performed per patch (2 from an area with UV coating and 2 from an area without UV coating), making a total of 32 tests. Test locations were prepared by coring the desired loca-

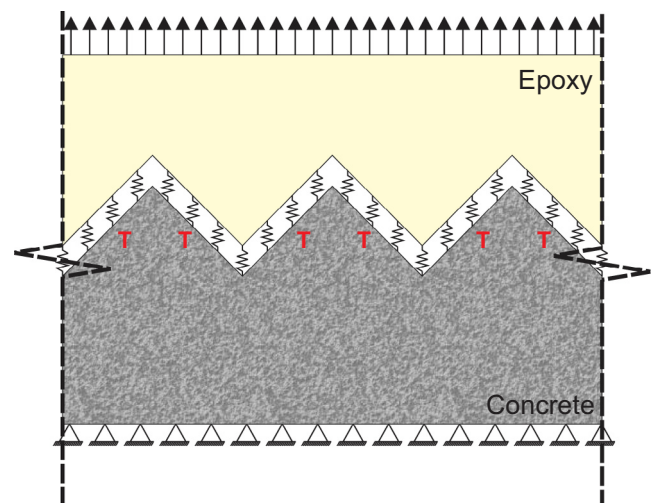


Fig. 11. Idealized representation of interface between concrete and epoxy loaded in Mode I: chemical bonds (shown as springs) are in tension (T); thus, the degradation of chemical bonds will directly affect the interface strength.

tion to approximately 0.25 in. deep, and adhering a dolly. The pull-off testing was conducted 24 h following the preparation of testing locations, to assure proper cure of the adhesive used to bond the dollies to the test patch. The pull-off test data is summarized in Table A2.

A total of 24 control tests were conducted by the contractor, within 24- to 48-h time window following the installation of FRP repair. Mean of all tests was 2.22 MPa with a COV of 0.08, with 19 of the total tests having a cohesive failure mode, while 5 exhibited the undesired failure mode through the dolly-adhesive interface

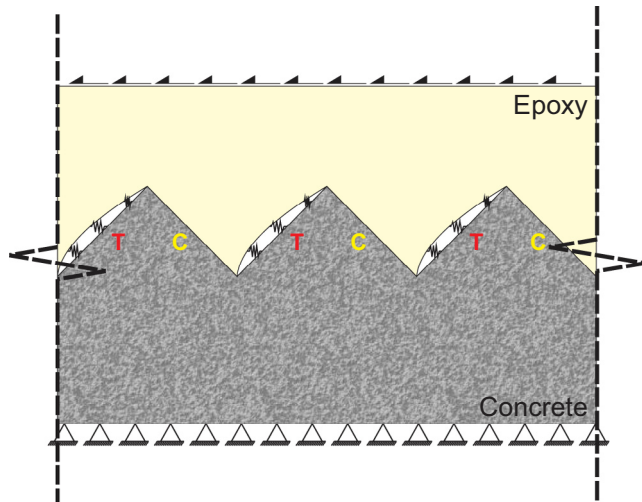


Fig. 12. Simplified representation of interface between concrete and epoxy loaded in Mode II: chemical bonds (shown as springs) are in tension ("T") or compression ("C"); thus, following the degradation of chemical bonds, the mechanical interlock is still a significant component in the total bond capacity.

(failure mode C, per ASTM D7522 [3]). All R_b values in Table A2 were calculated relative to the obtained mean value of control tests.

Per ACI 440.2R [5], for a test to be considered passing, the stress at failure should be a minimum of 1.38 MPa, followed by a cohesive failure. Other failure modes, such as adhesive failure mode are not explicitly discussed in the design guidelines; they are rather left up to "licensed design professional for evaluation and acceptance". Direct pull-off test results suggest that all tests satisfied the strength criterion, with an exception of a single test location (Table A2) on the exterior girder that was tested 6 months following the installation of the test patch.

Fig. 13 presents calculated R_b values from direct tension pull-off tests: large circular markers present the mean value of all tests for a specified test group with error marks representing one standard deviation; square markers correspond to individual test results. It should be noted that the majority of R_b values are greater than 1. Given that control tests were executed 24–48 h of initial installation of FRP, it is believed that the level of cure of the epoxy matrix has not reached its plateau, as for the 6 and 18 month testing marks. The spread of R_b data is quite large (Fig. 13); coefficients of variation in the test groups range from 0.20 to 0.40.

While within 6 months of exposure it is not clear if there are any significant differences in durability of bond between patches installed on interior and exterior girder, at 18 months what appears to be a clear difference between R_b values obtained from interior and exterior girder exists, with exterior girder having slightly lower R_b values. This is likely due to the more severe exposure conditions at the exterior girder, such as UV radiation and rainfall that are typically not as harsh at the interior girder.

More compelling evidence of increased degradation of FRP-concrete bond in the exterior girder is observed when comparing the failure mode data between the two girders (Fig. 14a). The presented results in Fig. 14 show an increase in adhesive failure mode characteristics from interior to exterior girder, with the difference being more apparent at 18-month testing mark; typical failure modes that were observed on the interior and exterior girder are shown in Figs. 15 and 16, respectively. A comparison of test data obtained from locations with UV coating to those without it, on the exterior girder (at 18 months of exposure), reveals the existence of correlation between the presence of UV coating and the failure mode. As shown in Fig. 14, adhesive failure mode characteristics increased at the location without the UV coating, indicating

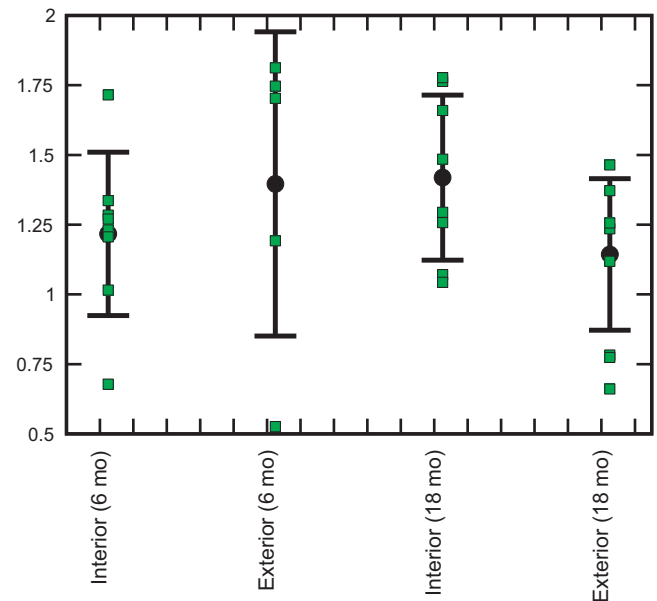


Fig. 13. Distribution of R_b amongst the test groups.

that UV radiation may be a significant contributing factor to the bond degradation. There was no evidence suggesting any dissimilarities in behavior between the patch without putty when compared to the one with putty; this could be likely due to inadequate sample sizes to allow for a meaningful comparisons between the two test groups. Effects of exposure to sunlight were observed on the surface of CFRP composite without UV coating in a form of scaling and change in color of the saturating resin; the interior girder patch locations without UV coating maintained the same color and glossy finish that was obtained following the initial cure of the composite.

8.2. Notched beam three-point bending tests

Results from notched beam three-point bending tests are summarized in Fig. 17 and Table A3. R_b values for notched beam three-point bending test appear to be consistent throughout the test groups; ANOVA analysis confirmed that no statistically significant difference exists between the test groups. Most R_b values were greater than 1.0; the likely explanation for this behavior is that in-situ environmental conditions allowed for higher cure in conditioned samples when compared to results from control specimens. Moreover, all conditioned laboratory samples were tested in a wet state, meaning that epoxy resin was not allowed any recovery, which would result in lower bond strengths. All field exposure samples were tested about 24 h upon collection from the dolphin, and were completely dry.

Analysis of fractured surfaces revealed no significant change in failure mode when compared to control, with an exception of samples that were exposed to UV radiation (Fig. 17b). It is expected that drying of bond line between epoxy and concrete had occurred in samples that were exposed to UV radiation, which preserved the epoxy adhesive properties, resulting in lower average percentage of adhesive failure.

9. Comparison of laboratory and field conditioning

9.1. Severity of conditioning environments

FRP-concrete bond was exposed to a variety of laboratory and field conditions to assess the appropriateness of ACP for estimation

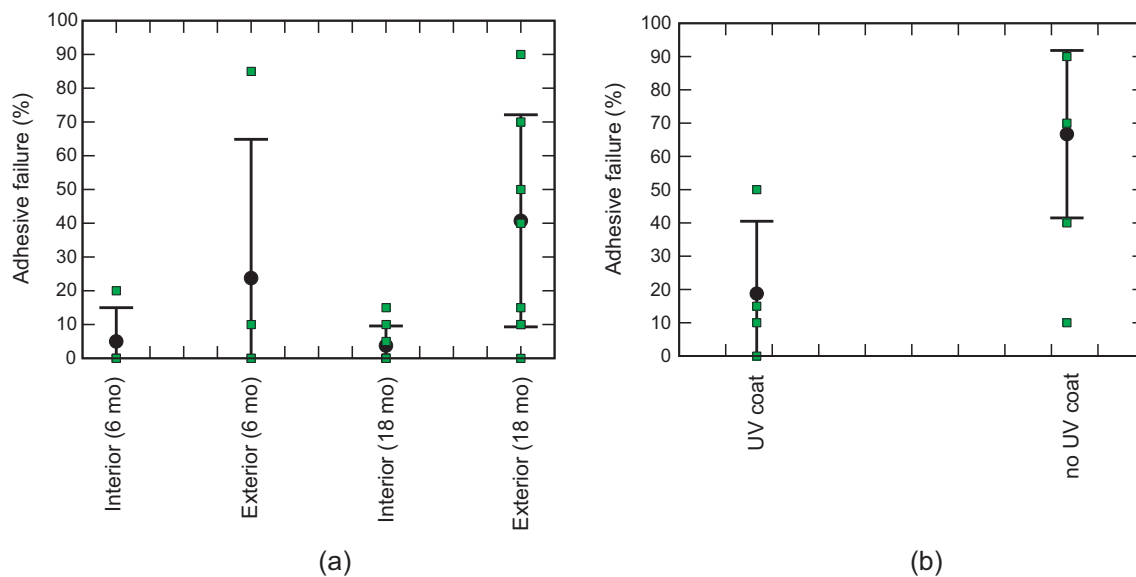


Fig. 14. (a) Distribution of adhesive failure modes amongst the test groups; (b) distribution of adhesive failure mode characteristics on exterior girder at 18 months of exposure.



Fig. 15. (a) Typical cohesive failure mode; (b) typical mixed failure mode.



Fig. 16. (a) Typical adhesive failure mode (system with putty); (b) typical adhesive failure mode (system without putty).

of durability of FRP-concrete bond. To aid in the comparison of the conditioning protocols, a ranking system of the exposure severity was developed (Table 3). Severity levels, ranging from 1 to 5, were assigned to each of the conditioning protocols used in this research. Data by Choi et al. [11], collected for the same composite following an 18-month exposure to brackish water environment on a fender system of a bridge on Matanzas River at Crescent Beach, Florida, were also considered. For comparison purposes, arbitrary severity levels were devised to allow ranking of the various exposure conditions. Average ultimate strength of tested specimens was plotted against severity of exposure in Fig. 18; a decrease in ultimate load with respect to exposure severity is evident.

To further investigate the effects of accelerated conditioning relative to field conditioning, laboratory 8-week exposure ultimate strengths were normalized to corresponding field conditioning ultimate strengths (Fig. 19). Lower normalized strengths indicate a more severe ACP. Interestingly, the average normalized strength for immersion at 30 °C was very close to the field conditioning. On the other hand, 60 °C conditioning generally produced lower ultimate strengths than when exposed to field conditioning. While 18 months of conditioning on the dolphin is more severe than that of the bridge girder due to its exposure to waves, spray and sun, it is not considered long enough to indicate with high confidence the durability during the service life of repair. Based on the results in Fig. 17a, however, there is evidence that conditioning of Composite

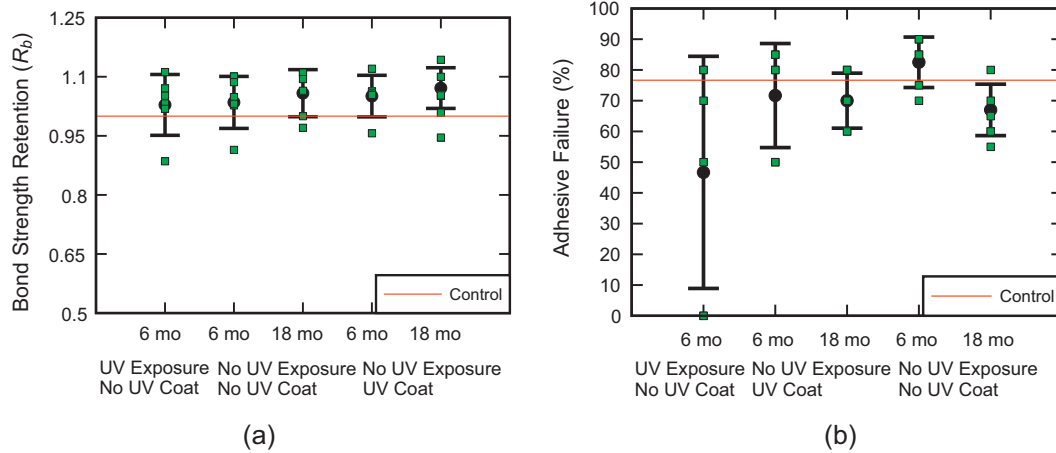


Fig. 17. Distribution of: (a) R_b ; and (b) adhesive failure modes, amongst the test groups.

Table 3
Ranking of exposure conditions according to severity level.

Severity level	Exposure time	Environment
1	18 months	Sunshine Skyway dolphin
2	18 months	Matanzas River bridge—brackish water tidal zone
3	8 weeks	Accelerated Conditioning by Water Immersion at 30 °C
4	8 weeks	Accelerated Conditioning at R.H. = 100% at 60 °C
5	8 weeks	Accelerated Conditioning by Water Immersion at 60 °C

A at temperatures of 60 °C may produce an overly pessimistic estimate of bond durability for the short (18 month) field exposure times used in this study. It is not clear whether the ACP will result in pessimistic results for longer field exposure times. It is expected that additional field testing will be conducted on the last set of samples at 10 years of exposure to provide long-term data for comparison.

No statistically significant difference in R_b values (Fig. 17a) over a 12 month period (between 6 and 18 months) of field exposure is a strong indication that either (i) the degradation of FRP-concrete

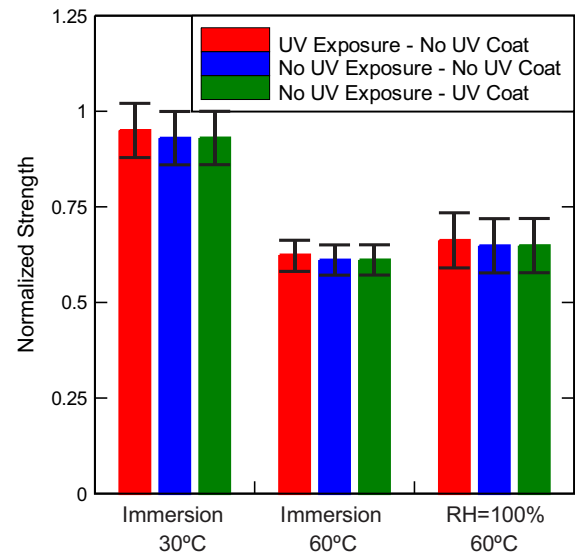


Fig. 19. Strengths of 8-week-exposure laboratory specimens normalized to strength of corresponding field conditioned specimens.

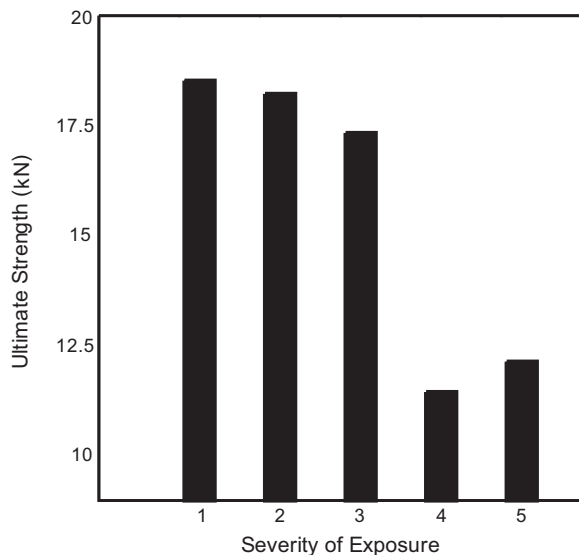


Fig. 18. Relationship between average ultimate strength and severity of exposure.

bond plateaued during the initial 0–6 months of exposure; or (ii) the rate of degradation in the field is extremely slow. One might expect differences in the bond strength of FRP under varying field exposure conditions. Fig. 18, however, shows insignificant difference between conditioning at the Sunshine Skyway dolphin and the more severe outdoor conditioning in the brackish water tidal zone [11].

9.2. Design considerations

The main purpose of three-point bending test results is to serve as a means to evaluate the potential durability of a bonded FRP system. In relation to the life-safety context of current ACI 318-14 Load Reduction Factor Design (LRFD) design specifications, Tatar and Hamilton [21] developed a method to experimentally characterize the lower bound on bond strength retention following exposure to ACP for the purpose of structural design [22]. The work was based on the results from accelerated conditioning of the notched beam three-point bending test specimens. Bond durability factor (BDF), as defined by the authors, represents 5% fractile of the experimentally obtained R_b values, which is considered a safe representation of bond durability from a design standpoint (ACI 318-14 [6]):

Table 4
Computed BDF values.

ACP	BDF
Water immersion at 30 °C	0.59
Water immersion at 60 °C	0.42
R.H. = 100% at 60 °C	0.29

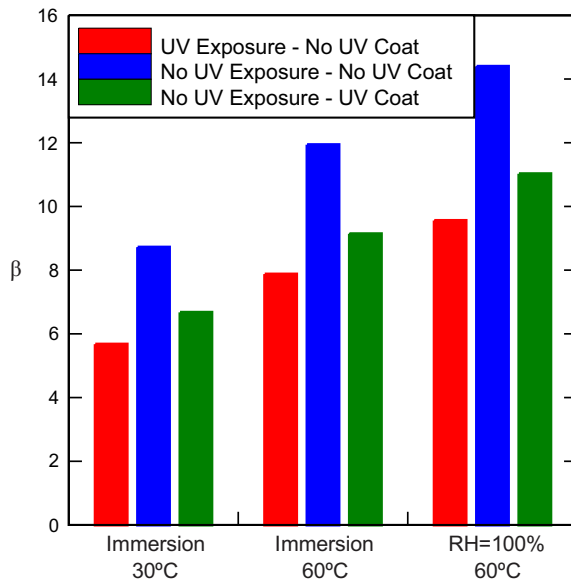


Fig. 20. β for different accelerated conditioning protocols.

$$BDF = \bar{R}_{b_{test,x}} \cdot (1 - K \cdot v_{test,x}) \quad (2)$$

where K – tolerance factor corresponding to a 5% probability of non-exceedance with a confidence of 90%, derived from a non-central t-distribution for which the population standard deviation is unknown. Values for specific samples sizes are provided in Hahn and Meeker [14]; BDF – characteristic BDF (5% fractile), lb/lb; $\bar{R}_{b_{test,x}}$ – mean R_b for test series x , where all samples in the test series have been exposed to same ACP, lb/lb; $v_{test,x}$ – coefficient of variation of the population sample corresponding to test series x , %.

Based on the comprehensive evaluation of durability data from notched beam three-point bending test [21], and those found in literature from other test methods [22], a BDF of 0.60 was identified as an appropriate value to characterize a lower bound R_b for wet layup systems that *do not* utilize putty (paste epoxy used to level the concrete surface). A similar value for systems that do utilize putty, such as Composite A, was not proposed due to their sensitivity to accelerated conditioning temperature (large discrepancy between conditioning at 30 vs 60 °C) discussed by Tatar and Hamilton [21].

BDFs from Table 4 were compared to the field test results, and expressed in terms of β (Fig. 20b), defined as:

$$\beta = \frac{\mu - BDF}{\sigma} \quad (3)$$

where μ and σ are respectively mean R_b and the corresponding standard deviation of the field test series, and BDF is determined for accelerated conditioning per Eq. (2).

β represents the number of standard deviations between the mean of 18-month-exposure field test data series and the corresponding BDF, and it can be used to specify design strength of materials used in infrastructure from a safety standpoint. For example, β used in specified concrete compressive strength ranges from approximately 1.3 to 1.5 (ACI 318 [6]). As shown in Fig. 20,

BDF values calculated from ACP specimens are significantly lower than those computed from field exposure data series; β ranges from approximately 6–15. Even though the long-term field exposure data are not available, it appears that K factor used to determine the BDF may be overly pessimistic, especially when paired with the exposure temperature of 60 °C. Long term field durability data (e.g. 10+ years of exposure) will provide experimental results that can be used to refine the proposed K factors.

Comparison of mean values for each of the Sunshine Skyway pull-off test series and those conducted in the laboratory setting reveals a large discrepancy between the two sets of test results. In the field, as discussed, due to the low control strength, mean R_b values were greater than 1; field test results were bound by those obtained from laboratory specimens for all exposure conditions. No further analyses were conducted in relation to pull-off test data.

10. Summary and conclusions

Accelerated conditioning is often utilized to estimate the durability of structural materials. FRP-concrete bond conditioning protocols typically include exposure to moisture and elevated temperatures. It is unclear, however, whether conditioning at temperatures higher than T_g of the adhesive provides a valid estimate of the bond durability or not. To help address this concern, this study presented a comparison of effect of field conditioning and accelerated conditioning protocols on the bond performance. Notched beam three-point bending and direct tension pull-off tests were performed on specimens that were aged under field conditions and in the laboratory to determine the applicability of laboratory ACP. The accelerated conditioning included immersion in 30 and 60 °C water while field exposure consisted of test patches on the approach span girders of the Sunshine Skyway bridge in Tampa Bay, Florida; at the same time, beam specimens were placed on the dolphins of the Skyway. The following conclusions were drawn based on the presented experimental evidence:

- All laboratory specimens showed a reduction in bond strength retention (R_b) throughout the exposure period, whereas the percentage of the bond area that exhibited adhesive failure mode increased. Sensitivity to temperature of bond between Composite A and concrete was noted—lower R_b values and higher average percentages of adhesive failure mode were observed in specimens conditioned at 60 °C, as opposed to those conditioned at 30 °C.
- No statistical correlation was found when comparing companion direct tension pull-off test results with three-point bending durability test results. Such behavior is accredited to the mechanics of adhesive joint failure under different loading conditions.
- As early as 6 months following installation, minor indications of FRP-concrete bond degradation were observed on the exterior girder CFRP patches compared to those applied to the interior girders. Reduction in bond capacity and increase in percentage adhesive failure when comparing the two locations suggests that the CFRP-concrete bond on exterior girder may have degraded due to UV radiation and rainfall.
- Comparison between laboratory and field data show that the exposure temperature of 60 °C may produce overly pessimistic estimates of bond durability for a short service life (less than 18 months). Eight-week exposure under all ACP, however, provided a lower bound on bond strength retention results for all field-conditioned beam samples.
- No change between the 6- and 18-month field exposure results for notched three-point bending specimens were noted indicating that the degradation of CFRP-concrete bond plateaus relatively quickly, or that the rate of degradation is extremely slow.

- In terms of the safety factor, BDF values determined based on laboratory durability data appear to provide an overly pessimistic estimate of bond durability when compared to 18-month dolphin field exposure. Conclusive evidence in regards to the conservatism of K factors used in BDF, however, remains inaccessible until long-term (10+ years) field data is available.

Appendix A

Table A1

Table A2

Table A3

Table A1

Summary of laboratory three-point bending test results.

Exposure Duration	Variable	Immersion at 30 °C	Immersion at 60 °C	RH = 100% at 60 °C
1 week	Strength (kN)	14.70	17.57	19.06
		15.13	16.97	18.17
		15.44	16.12	19.59
	Average (kN)	15.09	16.89	18.94
	COV (%)	2.5	4.3	3.8
2 weeks	Strength (kN)	17.42	13.92	14.35
		19.31	13.91	13.16
		15.86	15.16	18.02
	Average (kN)	17.53	14.33	15.17
8 weeks	Strength (kN)	9.8	5.0	2.53
		16.00	12.21	10.56
		18.59	10.77	12.73
	Average (kN)	17.33	11.39	12.08
Control	Strength (kN)	Standard Laboratory Conditions (SLC)		
			19.34	
			17.22	
			16.64	
	Average (kN)		17.74	
	COV (%)	7.5	6.5	10.9

Table A2

Summary of field direct pull-off test results.

Location	UV coating?	6 months			18 months		
		Stress at failure (MPa)	R_b	Failure mode (% adhesive)	Stress at failure (MPa)	R_b	Failure mode (% adhesive)
1-I-#1	Y	2.85	1.28	Dolly	3.29	1.49	0
2-I-#1	Y	2.82	1.27	Dolly	2.87	1.30	0
3-I-#1	N	1.50	0.68	0	3.68	1.66	15
4-I-#1	N	2.25	1.02	Dolly	2.79	1.26	5
5-I-#2	Y	2.96	1.34	0	3.91	1.76	10
6-I-#2	Y	2.73	1.23	0	3.94	1.78	0
7-I-#2	N	2.67	1.21	FRP	2.37	1.07	0
8-I-#2	N	3.80	1.72	20	2.31	1.04	0
9-E-#1	Y	1.17	0.53	100	1.74	0.78	10
10-E-#1	Y	3.77	1.70	100	2.48	1.12	0
11-E-#1	N	N/A	N/A	N/A	2.74	1.24	40
12-E-#1	N	N/A	N/A	N/A	1.47	0.66	70
13-E-#2	Y	2.64	1.19	90	3.04	1.37	15
14-E-#2	Y	4.02	1.81	15	2.78	1.26	50
15-E-#2	N	3.87	1.75	Dolly	3.25	1.46	90
16-E-#2	N	N/A	N/A	N/A	1.72	0.77	10

FRP: cohesive failure through FRP.

Dolly: interfacial failure between the adhesive and dolly.

N/A: test was not conducted.

I: interior girder.

E: exterior girder.

Table A3

Summary of field three-point bending test results.

Exposure Duration	Variable	UV Exposure/no UV coat	No UV exposure/no UV coat	No UV Exposure/UV coat
6 months	Strength (kN)	18.99	19.53	17.21
		19.72	19.27	18.88
		1.31	16.22	17.74
		17.07	18.60	19.69
		15.71	18.22	19.70
		18.66	18.30	19.41
	Average (kN)	18.24	18.36	18.77
	COV (%)	7.5	6.4	5.6
	18 months	Strength (kN)	N/A	16.77
			N/A	18.68
			N/A	18.85
			N/A	20.28
			N/A	18.76
			N/A	16.97
	Average (kN)	N/A	18.70	18.67
			18.64	18.63
	COV (%)	N/A	5.0	6.6

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